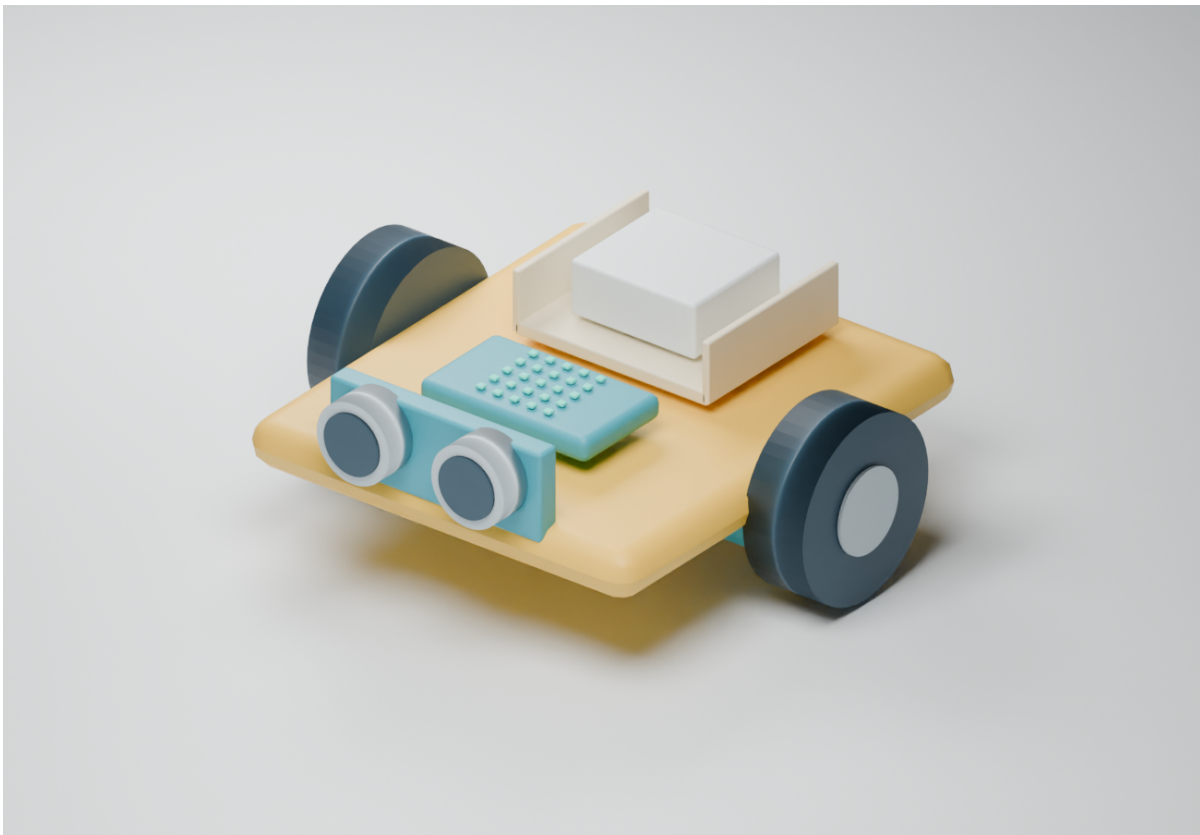


Build your first delivery rover

Student engineering workbook

Grades 6-8 • 8 modules • 16 sessions of 45 minutes

BBC micro:bit V2 + Kitronik :MOVE Motor (5683)



Conceptual Blender illustration. Use the manufacturer guide for actual assembly.

Predict. Build. Test. Explain. Improve.

Build a robot that can deliver a small paper parcel, detect an obstacle, and stop at its destination. Work like an engineer: predict, build, test, explain, and improve.

Your engineering agreement

Name: _____ Team: _____

Our goal

Carry a small paper parcel and stop in a delivery bay before a cardboard obstacle. Every lesson adds a tested capability to the same rover.

Work safely

Start on a clear floor with an adult ready to switch off the robot. Button A starts a trial and button B requests a software stop. The physical power switch is the immediate stop if code or controls fail. Reference programs also have a short time limit. Test each program on the actual kit before class; wheel motion and stopping distances require local calibration.

Students rotate driver/programmer and observer/data-recorder roles after each trial. Keep hair, fingers and loose clothing away from wheels. Switch power off before adjusting the robot; keep liquids and metal objects away from the board. Follow the school's and manufacturers' safety instructions.

Before every run

Clear floor lane. Secure cargo. Sensor unobstructed. Known program. Driver and observer ready. Teacher approval. Physical switch accessible.

Keep useful evidence

Record unsuccessful trials too. Change one factor at a time. Keep the start position, floor, load and program fixed within a trial set.

Before we begin

One robot I know, its job, and an input and output it uses:

Our team stop procedure and roles:

Sense, decide, respond

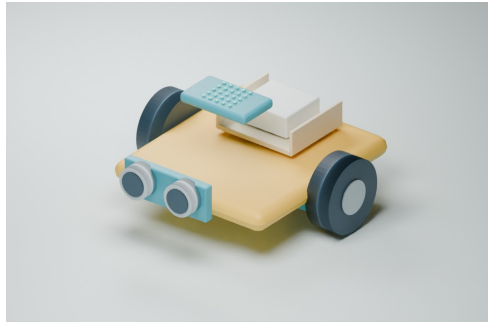
Build a rover that responds to its operator

Your target

Identify an input, a programmed decision and an output, then demonstrate a predictable response in three trials.

Gather

Assembled rover, micro:bit, USB cable and computer. Keep motor power off until the teacher checks the program.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

A robot is a programmable physical system that acts on its environment. Some robots are remotely operated; others use sensor readings to act autonomously. In this course, sense–decide–act is a useful design model, not a rule that excludes every remotely controlled robot.

Build and test

1. Draw the rover and label the micro:bit controller, a button input, the ultrasonic sensor, motors and wheels. Show the direction of information flow.
2. Open lesson 1 in MakeCode. Predict the micro:bit display after A and after B. This program deliberately keeps both motors stopped.
3. Download and test on the mounted controller. Press A, then B; the display changes from a tick to a cross. Treat this as the rover's operator interface.
4. Change the ready display to your team's original five-by-five icon. Keep the stop response. Explain which part of the program interprets the button event.

Success criteria

A shows the chosen ready icon, B shows the stop icon, and wheels remain still in all three trials.

Use starter program: `code/lesson-1.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Input: button A or B. Observe: display and wheel state. Keep the same program between trials.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / button	Predicted output	Observed output	Matches?
1			
2			
3			

Explain and improve

A remotely operated exploration robot is still a robot. What extra capability would make this rover autonomous?

My evidence supports this change / next test:

Motor checkout

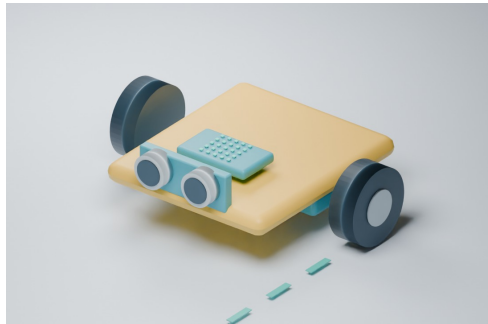
Make a controlled, repeatable drive

Your target

Run a bounded forward drive, demonstrate stop, and identify two sources of variation.

Gather

Rover, ruler, floor tape and clear 1.5 m test lane. Teacher first tests with wheels clear of the floor, holding only the chassis.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

Each drive wheel has a motor. Two similar wheel speeds move the rover approximately straight; unequal speeds create a turn. A speed setting is a motor command, not a measurement in centimetres per second.

Build and test

1. With power off, identify both motors and the physical switch. Check wheels are secure and nothing rubs. Do not disconnect factory wiring.
2. Mark a start line. Set the reference program to speed 25 and drive time 800 milliseconds. These are starting values, not calibrated distances.
3. Place the rover behind the line. Press A from a safe position. Observe the automatic stop; record travel distance. Repeat three times without changing the code.
4. Run again and press B during motion. Verify the software stop. Keep the power switch accessible; it is the immediate fallback.
5. If travel curves, first check the surface, wheels and batteries. Change only one cause or setting, then repeat.

Success criteria

Three automatic stops and a demonstrated B stop, with travel distance and variation recorded.

Use starter program: `code/lesson-2.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change only one factor, such as commanded speed. Keep time, start direction, floor and load fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / speed setting	Time (ms)	Distance (cm)	Drift / stop notes
1			
2			
3			

Explain and improve

Why might speed 25 produce a different distance tomorrow?

My evidence supports this change / next test:

Design a cargo cradle

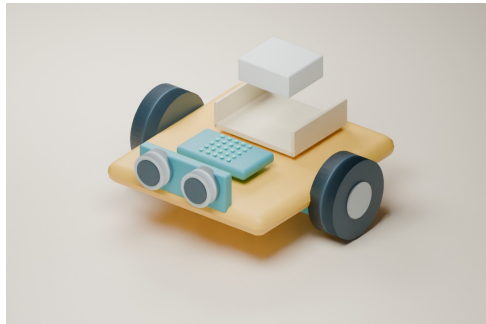
Carry a paper parcel without blocking the robot

Your target

Build and compare two removable cargo cradles using consistent trials and a stated design constraint.

Gather

Rover, folded paper parcel, lightweight cardboard, reusable adhesive, ruler and scissors. Teacher approves attachment before driving.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

A chassis supports components. An attachment changes how weight is distributed and may obstruct sensors or wheels. Engineers compare designs against constraints, rather than assuming that a stronger-looking structure performs better.

Build and test

1. Make the same lightweight folded-paper parcel for both designs. Do not add metal weights. Sketch a low cradle that leaves the ultrasonic front, underside sensors, wheels, buttons and power switch clear.
2. Build a removable cradle on the available upper mounting area, following manufacturer restrictions. Do not tape across circuitry, wiring or sensor faces. Teacher checks the fit with power off.
3. Use lesson 3's short drive. Test the same distance/time setting three times. Record whether the parcel stays in place and whether the attachment rubs or tips.
4. Revise one feature, such as the height of a paper retaining wall or cradle position. Repeat on the same floor with the same parcel and code.
5. Choose a design using your results. Draw its attachment points so another team can reproduce it.

Success criteria

Parcel stays in place for three short trials; the sensor view, wheels, controls and switch remain unobstructed.

Use starter program: `code/lesson-3.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change one cradle feature. Keep parcel, surface, start position and program fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Design / trial	Parcel retained?	Any rubbing or tipping?	Evidence / revision
1			
2			
3			

Explain and improve

Why should you keep the parcel and code unchanged when comparing the cradles?

My evidence supports this change / next test:

Give the rover reliable senses

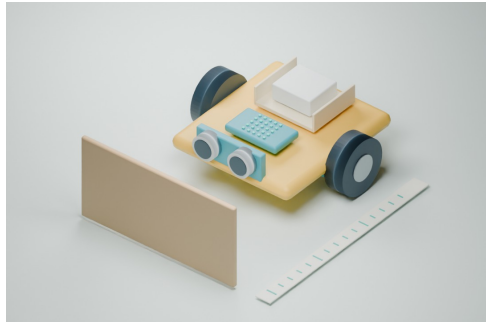
Build and calibrate a parking-distance display

Your target

Compare distance readings at three measured positions and explain one unreliable reading.

Gather

Rover, ruler, upright flat cardboard wall and test lane. Motors stay off for this activity.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

An ultrasonic sensor estimates distance from an echo. Readings can be affected by object angle, shape and surface. A missing or invalid reading is not evidence that the route is clear. Calibration compares readings with a reference measurement.

Build and test

1. Keep wheels stationary. Use the manufacturer's centimetre unit setting in lesson 4. The micro:bit shows the measured distance; display scrolling is acceptable here because the rover is not driving.
2. Measure from the front sensor face to the cardboard wall. Place it at 10, 20 and 30 cm. At each location, take three readings and record the variation.
3. Tilt the wall or substitute a soft object. Observe whether readings change. Do not point the sensor at faces or use people as obstacles.
4. Choose a warning threshold such as 20 cm for a later low-speed approach. Explain why real braking distance must be tested separately.
5. Trace the information path: distance sensor → controller comparison → display now, motors later.

Success criteria

Nine readings recorded; measured and reported distances compared; one limitation explained.

Use starter program: `code/lesson-4.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

First vary measured distance only. Then vary the target angle while holding distance fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Measured distance (cm)	Readings 1 / 2 / 3	Difference / variation	Surface or angle
10			
20			
30			

Explain and improve

The sensor returns zero. Should your moving robot assume there is no obstacle?

My evidence supports this change / next test:

Code a delivery route

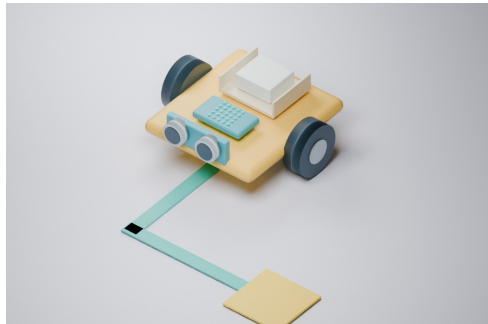
Drive to a marked delivery bay

Your target

Program a two-leg route and use three trials to tune one timing variable.

Gather

Rover, empty cargo cradle, floor tape and ruler. Mark a generous delivery bay, starting at least 30 cm from any obstacle.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

A sequence runs commands in order. Timed movement is open-loop: the program does not check where the rover actually ended up. A turn duration is not a guaranteed angle. Calibrate movement on the same floor used for the mission.

Build and test

1. In the online Mission Lab, predict a route before running it. Try forward, turn and stop commands. The grid is a logic model, not a physical calibration tool.
2. On the floor, mark a start, a turn point and a delivery bay. Sketch the intended route with arrow directions.
3. Open lesson 5. Start with short forward times and a low-speed timed spin; test the turn alone before combining the route. A 400 ms spin is only an initial tuning value.
4. Run the two-leg route three times. Measure how far the stopping point is from the bay centre. Change only turn time or one drive time, then retest.
5. Place a repeat block around a short segment only if your chosen route repeats. Explain why blindly repeating an inaccurate segment compounds error.

Success criteria

A documented two-leg route ends within the team's stated bay tolerance in at least two of three trials, and stops every time.

Use starter program: `code/lesson-5.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change one timing value. Keep speeds, floor, start alignment and load fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / timing change	End error (cm)	Turn observation	Inside bay?
1			
2			
3			

Explain and improve

Why can a route succeed once but miss the bay on the next run?

My evidence supports this change / next test:

Stop before an obstacle

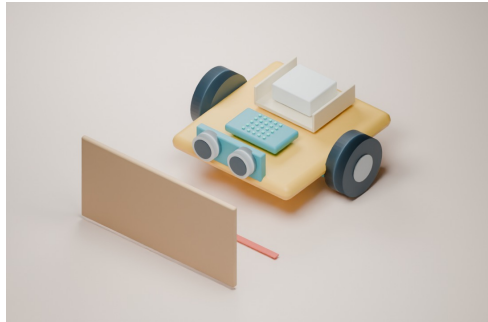
Build a cautious autonomous approach

Your target

Demonstrate a sensor-controlled stop before contact in three trials, then explain the threshold using data.

Gather

Rover, wide upright cardboard wall, ruler and clear floor lane. No people, animals, stairs or table edges as targets.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

A conditional chooses an action from sensor evidence. Your rover repeatedly compares distance with a threshold. It stops on a close or invalid reading. Sampling, movement and braking take time, so a threshold must include a tested margin.

Build and test

1. Start at least 50 cm from the cardboard wall. Use low speed and an initial threshold of 20 cm. Teacher pretests and increases the threshold if needed.
2. Predict which action occurs for distances of 30 cm, 15 cm and zero. Test that reasoning with the Mission Lab's "if blocked" command.
3. Load lesson 6. Press A. The rover approaches, samples distance repeatedly and stops when the reading is at or below the threshold or invalid.
4. Measure the actual remaining gap after each of three trials. Record any contact as a failed test. Do not hold the target by hand during motion.
5. Improve one parameter. If stopping is late or inconsistent, lower speed and increase the margin, testing changes separately. Keep the invalid-reading stop.

Success criteria

Three stops before contact, each with a measured gap, plus an explanation of one failure or limitation.

Use starter program: `code/lesson-6.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change threshold OR speed, never both in the same comparison.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / threshold (cm)	Speed command	Final gap (cm)	Contact / observation
1			
2			
3			

Explain and improve

Why might increasing the stopping threshold help a rover that reacts too late?

My evidence supports this change / next test:

Plan the delivery mission

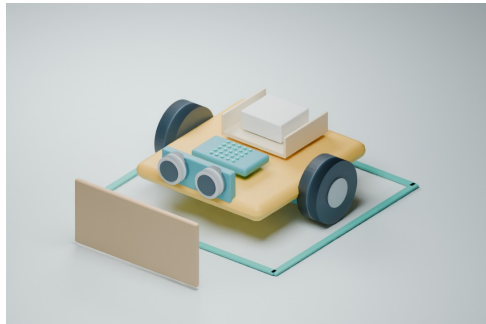
Deliver a paper parcel using a tested approach

Your target

Combine a cargo attachment, a bounded route and a sensor stop into a mission with measurable acceptance criteria.

Gather

Rover, approved paper cradle and parcel, floor bay, flat cardboard stop target, ruler and worksheet.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

A mission combines requirements, constraints and evidence. Reliable behaviour is more valuable than speed. A state is a named stage such as ready, approaching or stopped; only one stage should control the motors at a time.

Build and test

1. Write a mission brief: carry the same paper parcel, follow a clear approach lane and stop in a marked bay before the flat target. Set the bay using the stopping gaps measured in lesson 6.
2. Assign rotating roles and sketch the lane. List hazards and the immediate stop method. Teacher approves the setup.
3. Use lesson 7's reference approach program and your measured speed and threshold. Start with a straight approach; add a timed turn only after testing it separately.
4. Run three trials with identical starting conditions. Record parcel retained, bay reached and no contact. A trial passes only when all three are true.
5. Choose one improvement from the evidence. Retest three times and compare success counts; keep unsuccessful results in your log.

Success criteria

At least two of three full mission trials meet all three criteria; every trial remains bounded and supervised.

Use starter program: code/lesson-7.ts. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Choose one deliberate change between trial sets, then keep it fixed within each set.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / version	Parcel retained?	In bay / no contact?	Pass? Evidence
1			
2			
3			

Explain and improve

Your first version passed 1 of 3 trials and your revision passed 2 of 3. What can you claim?

My evidence supports this change / next test:

Showcase with evidence

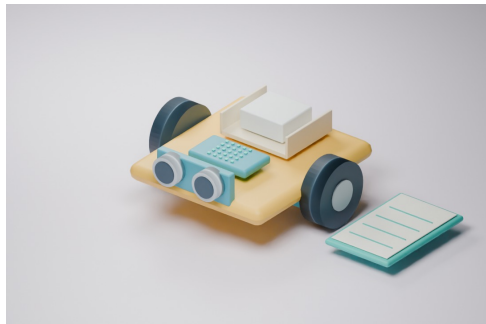
Prove, explain and hand over your robot

Your target

Present a reproducible demonstration and defend a design choice using measurements and a limitation.

Gather

Final rover, unchanged mission layout, saved program, engineering log and presentation worksheet.



Concept illustration; use the manufacturer guide for actual assembly.

Understand

An engineering handover lets another person understand, operate and evaluate a design. Clear documentation includes the actual limits and remaining problems. A demonstration shows one run; a test log shows what happened across runs.

Build and test

1. Freeze a named program version and draw the final layout. Write setup, start, stop and reset instructions for another pair.
2. Run three final acceptance trials with the same parcel and target. Record all results before selecting a demonstration clip or photo.
3. Have another pair follow your instructions under teacher supervision. Observe unclear steps and revise the handover without changing the test data.
4. Give a two-minute presentation: need and criteria; sense–decide–act explanation; one design change with before/after evidence; one limitation and next test.
5. Complete a peer review and individual reflection. Save the MakeCode project and export the worksheet or log. Teacher assesses each student's explanation as well as team performance.

Success criteria

A reproducible handover, three recorded trials, a two-minute evidence-based explanation and an individual reflection.

Use starter program: `code/lesson-8.ts`. Add Kitronik's extension in MakeCode first.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Keep the final design fixed during acceptance tests. Label any later change as a new version.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Final trial	All criteria passed?	Observed failure / gap	Next test
1			
2			
3			

Explain and improve

What should you say if the showcase demonstration fails?

My evidence supports this change / next test:

Final mission handover

Program version / saved project: _____

Our mission criteria and layout

Setup → start → stop → reset

Two-minute showcase

0:00-0:20: need and criteria. 0:20-0:50: input, decision, output. 0:50-1:30: before/after evidence. 1:30-2:00: limitation and next test.

Peer review

One instruction I could follow: _____

One instruction that needs clarification: _____

Individual reflection

A choice I made, the evidence I used, and what I would test next:
